

Vision Statement

Nursing students often enter clinical practice with strong theoretical foundations but minimal hands-on experience navigating digital health systems. When these students transition to clinical settings and encounter an Electronic Medical Record (EMR) software, this inexperience can cause inefficiencies and potential errors. Our project, Calvin EMR, addresses this challenge by creating a realistic training platform specifically designed for our Nursing program students. Building upon the foundation established by previous students, our team will expand the system's capabilities by adding features as requested by the Nursing department, while developing a comprehensive collection of end-to-end tests. This will enable the students to gain practical experience with patient data management and EMR workflows in a risk-free environment. Beyond feature development, we are committed to keep our code base clean, testable, scalable, and maintainable. Our ultimate goal is to develop a high quality educational tool that helps Nursing students learn about digital medical systems, while also establishing a foundation that can be adapted for potential future commercialization.

Normative and Ethical Considerations

Appropriateness

Technology products should align with the culture and practices of their intended context, fostering meaningful improvement while respecting established workflows and professional standards.

Consequences if ignored: The simulation may omit important features that nursing students actually need or it may use incorrect medical terminology.

Remedy: We work closely with the Nursing department to ensure we include the right features and use proper clinical language.

Stewardship

Use of creational resources should be respectful, frugal, and caring; concern for sustainability, limits, and the environment.

Consequences if ignored: Poor code quality and lack of documentation will burden future student teams and waste institutional resources.

Remedy: We prioritize clean code, clear documentation, and thorough testing to ensure the project remains maintainable.

Caring

Our designs should help us serve one another, promote wellness, contribute to healing, show love to our neighbor, and enable fellow creatures to flourish.

Consequences if ignored: We may end up making the patients feel more like data points to the nurses rather than actual people (even if it's a fake simulation).

Remedy: Make patient data feel personal by including profile pictures and care notes that remind students the data are real individuals.

Trust

We aim to build a system that behaves consistently and communicates honestly, so there is no ambiguity about any process or workflow.

Consequences if ignored: Without proper communication from the system, the users might be left in ambiguity about the actions performed, losing trust in the system.

Remedy: Provide real-time feedback about certain actions performed, and write tests to ensure predictable behavior from the system.

Product Success Criteria

Since this project builds upon previous student work, we will measure success by:

- Maintaining existing functionality while adding new features
- Creating comprehensive automated tests that ensure no regressions occur
- Engaging regularly with the nursing department, directly or through our advisor, to validate that our implementation aligns with their needs

Development Process

Our team has established an effective development workflow using GitHub Flow. We have forked the main repository and we develop features on dedicated branches, and submit Pull Requests to the staging branch for review before merging. We meet weekly with our advisor to review progress and coordinate upcoming work. Each team member commits 6 hours per week to the project.

Development Environment

All team members have configured their development environment with the required technology stack: Node.js, Git, Angular, Typescript and Firebase. All members have verified that they can

run the application locally, execute tests, verify changes with testing credentials, and follow GitHub Flow. Everyone is equipped with the necessary tools, access, and test accounts to be productive on the project.

Team Contract

Time commitment: 6 hours per week

Meetings: Weekly meeting with our advisor, and additional team meetings as required

Responsibilities: Each member owns specific features/issues and assists teammate as needed

Communication: We will use Teams for updates to coordinate meeting times

Code Standards: All code must pass code review from our advisor before merging

Updates Since Milestone 1

What We Accomplished So Far

Since milestone 1, our team has made meaningful progress in terms of new features and bug fixes. We have implemented contextual pop-up notifications to provide users with real time feedback on various actions, including saving care plan notes and creation of new patients, students, and medications. The goal of these notifications is to improve overall user experience by confirming that operations have been completed successfully.

We have also resolved several bugs that were affecting the system usability. The “select all” control on the patient list was fixed and now behaves correctly. We also added the functionality of automatic list refresh when creating a new patient, so the record appears immediately without requiring manual refresh. Additionally, we fixed a logic error in the “within normal limits” (WNL) feature for patient assessment that was producing incorrect results.

Individual Contribution to Codebase

Swayam

- Implemented pop-up notification functionality across various operations to provide real-time user feedback
- Currently updating multiple forms to include validation for required fields, making them more visible and intuitive
- Updating patient creation tests to cover the new validation logic

Alvin

- Resolved the broken "select all" functionality on the patient list
- Implemented automatic list refresh so newly created patients appear without a manual reload
- Currently working on SAML login integration with the EMR system to enable sign-in using Calvin University email credentials

Peter

- Fixed the WNL bug that was not behaving correctly for patient assessment
- Currently developing unit tests for WNL assessments to ensure continued correctness

Presentation Outline

1. What is an EMR?
 - Overview and benefits for nursing student training
2. Project Overview
 - Goals and planned additions
 - Testing strategy
3. Updates we have made
 - Features added and bugs fixed
4. Design Process
 - Collaboration with nursing department to guide priorities
5. Live Demo
6. Future Plans

Results:

* = required

Error
Please fill out required fields before submitting.

— Identification

First Name *Required ** Middle Name Last Name *Required **

Medical Record Number *Required ** Generate MRN

— General Information

Type *Required ** Age (years) *Required ** Sex *Required **

Date of Birth
Month *Required ** Day *Required **

Weight (kilograms) *Required ** Height (feet) *Required ** Height (inches) *Required **

Code Status *Required ** Isolation *Required **

Cardiovascular

WNL: Regular rate, rhythm, SI S2, no edema, extremities warm with capillary refill less than 3 seconds. Pulses strong. No chest pain or palpations. No cyanosis.

All Within Normal Limits

Heart Tones

Pulse Pattern

☐ Irregular

Heart Tones

WNL

Rhythm

PVC's

Edema

☐ Presence of JVD

☐ Generalized Non-Pitting Edema

Pitting Edema

Right Hand

1+

Left Hand

WNL

Right Foot

3+

Left Foot

WNL

Updates Since Milestone 2

The Design Norms section was updated to include 'Trust', and screenshots for the feature updates have been added in the results section. The project progress is as expected and is going well. We have been able to implement the committed features and closely follow the Product Success Criteria.